INCH-POUND

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MILITARY SPECIFICATION

RADIATORS, ENGINE COOLING, INDUSTRIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers coolant radiators for liquid-cooled, industrial, internal-combustion engine of 10 horsepower and above.

1.2 Classific tion. The radiators shall be of the following types, as specified (see 6.2):

Type I	- Soldered-tank radiator.
Type 11	- Bolted-tank radiator with replaceable core.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, Attn: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD) Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2930

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SPECIFICATIONS

FEDERAL

W-L-800	- Lubrication Oil, General Purpose, Preserative (Water-Displacing, Low Temperature).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Presentation, Methods of.
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal.
MIL-G-12803	Gasket Material, Non-Metallic.
MIL-T-22085	- Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.
HS35773	- Radiators, Engine Cooling, Industrial.

(Unless othewise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 308	Steel Sheet, Cold-Rolled, Long Terne Coated.
B 36	- Brass Plate, Sheet, Strip, and Rolled Bar.
B 152	Copper Sheet, Strip, Plate, and Rolled Bar.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description.

3.1.1 Type I. The type I, soldered-tank radiator, shall be as shown on MS35773, dash numbers -1, -2, -3, -9, -10, or -11 and as specified herein.

3.1.2 Type II. The type II, bolted-tank radiator with replaceable core, shall be as shown on MS35773, dash numbers -4, -5, -6, -7, or -8 and as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. The material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subjected to all provisions of this specification.

3.3.1 Material deterioration Prevention and control. The radiator shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the radiator may be exposed.

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3.3.1.1 Dissim- metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the radiator may be newly fabricated from recovered materials to the maximum extent practicable, provided the radiator produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the radiator.

3.3.3 Copyer tubing. Tubes shall be copper, red brass, or naval brass. Tubing shall not be subject to dezincification and resultant corrosion.

3.3.4 Sheet Sheet copper for fins shall conform to ASTM B 152.

3.3.5 Sheet brass. Sheet brass for tanks and header plates shall conform to ASTM B 36.

3.3.6 Terme plate. Terne plate for side members for type I radiators shall conform to ASTM A 308 Terne Plate, with LT35 coating thickness as a minimum.

3.3.7 Gasket. Gasket material shall conform to MIL-G-12803, type 1, identification number F112600E56M6.

3.4 Fins. The core assembly shall be of the tube and plate-fin construction with the fins perpendicular to and in close contact with the tubes.

3.5 Peformance.

3.5.1 Heat rejection and core resistance. The heat rejection and core resistance curves of the test core shall be compared with the corresponding heat rejection and core resistance curves of the standard core. A fan curve at standard air density (0.075 lbs/cu. ft.) shall be drawn through the standard core resistance points at the 1500, 1800 and 2100 feet per minute air velocity points. The point at which the fan curve intersects the test core resistance curve shall be projected onto its respective heat rejection tune. When compared to the standard core values, the projected core values shall conform to the values stated on sheet 4 of MS35773.

3.5.2 Heat rejection compariSeq. The comparison between heat rejection (air gain) and heat rejection (water loss) shall not exceed 5 percent.

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3.5.3 Distortion. When tested as specified in 4.5.2.3, the radiator and core shall show no leakage or permanent distortion exceeding 1/8 inch.

3.5.4 Vibration and leakage. The radiator or core when vibrated as specified in 4.5.2.4.2 at the most critical resonant frequency shall show no evidence of structural damage, seepage, or leakage.

3.6 Identification marking. Radiators and cores for type II radiators shall be identified in accordance with MIL-STD-130. A metal identification plate shall be permanently affixed to the side of the top tank of each radiator near the water inlet. In addition, a metal identification plate shall be permanently attached to the side of the core for the type 11 radiator. Marking shall include the following information:

Military standard radiator
MS part nuber
Manufacturer's name or trademark

3.7 Finishing and painting. The radiator and core shall be finished and painted in accordance with the contractor's standard practice.

3.8 Workmanship.

3.8.1 Cleaning. The completed radiator and cores shall be cleaned and free from defects such as dirt, sand, metal chips, rust, cracks, and other foreign materials or defects that could impair their serviceability. Water from the cleaned radiator or cores shall not show a "pH" change of greater then 1, the chloride content shall not exceed 75 parts per million, nor shall there be a change in chloride content no greater than 75 parts per million when compared with water before rinsing.

3.8.2 Soldering. Soldering of the overflow tube to the filler neck and the filler neck to the radiator top tank shall not interfere with the proper operation of the pressure cap. In the event any soldering is done to parts after the radiator or core has been cleaned, the radiator or core shall be cleaned again after the soldering operation.

3.8.3 Header plates. The gasket area, the corresponding bolt-holes, and the bolt head area of the heater plates used on type II radiators shall be free of solder lumps and a water resistant nonhardening cement shall be used on the gaskets when the core is assembled to the tanks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Conmonent and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, l s applicable.

4.1.3 Disassemblv inspectiom. Failure of any test by the first article model shall be cause for disassembly, in the presence of a Government representative, of the first article model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection. First article inspection shall be performed on a preproduction model. The first article shall consist of one or more units (see 3.2 and 6.3).

4.3.1 Examination The first article radiator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article radiator shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4-1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

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4.4-2 Examination. Sample selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The radiator shall be examined as specified herein for the following defects:

101. Dimensions not as specified (see 3.1).
102. Materials not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable operating and storage environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 ae not effectively insulated from each other (see 3.3.1.1).
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.1.2).
106. Used, rebuilt or remanufactured components, pieces, and parts incorporated in the radiator (see 3.3.2).
107. Core l ssembly not as specified (see 3.4).
108. Identification marking incorrect or illegible (see 3.6).
109. Finishing and painting not as specified (see 3.7).
110. Workmanship not as specified (see 3.8).

4.5.2 Tests.

4.5.2.1 Conditions. The following conditions shall apply during the tests performed in accordance with this specification on complete radiators and on radiator cores when they are furnished separately. Radiator cores furnished separately shall be tested in fixtures which simulate, for test purposes, the top and bottom radiator tanks.

4.5.2.1.1 Supports. The radiator or core shall be supported on its normal points of support and shall not be supported on a cradle or bracket which in any way restrains the possible distortion of the radiator or core when under pressure.

4.5.2.1.2 **Equipment**. The test radiator or core, as applicable, shall be mounted on a test stand. A circulating pump shall be used to maintain an even temperature in the water resevoir. A flow pump and piping shall be used to provide water to the test radiator or core and a throttling valve shall be used to regulate the flow to the desired rate. A variable speed fan or adjustable dampeners or louvers shall regulate the airflow to the test radiator or core.

4,5.2.1.3 Instrumentation. Instrumentation shall be provided to perform the following functions:

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a. To measure the temperature in the waterline at inlet and outlet to test radiator or core.
b. To measure airflow in the air duct. (When a orifice is used to measure the airflow, a manometer shall be connected upstream from the orifice to indicate the static pressure in the duct.)
c. To measure the pressure drop or the resistance to airflow across the core.
d. To measure the temperature of the airflow on each side of the core.

4.5.2.1.4 Control limits and data observations. The observed data shall be recorded. All points for each test shall be recorded only after all variables have been stabilized. The degree of stabilization and accuracy of observations shall be acceptable if the heat-rejection comparison conforms to 3.5.2.

4.5.2.1.5 Coolant. The coolant shall be water.

4.5.2.1.5.1 Coolant temperatures. The temperature of the water entering the test section shall be between 170 and 210 'F.

4.5.2.1.6 Heat rejection. The heat rejection test shall be made using one of the following test methods.

4.5.2.1.6.1 Heat rejection. test me thod (a). The core section shall be tested in the wind tunnel at the 100 percent water flow rate and at 125 percent of the rated water flow (see 6.4.4). If these selected values are not possible the core shall be tested at not less than three water flow rates which bracket the 100 and 125 percent rate for each radiator size. At each water flow rate, the heat rejection shall be determined at not less than four air velocities overlapping the range of 1500 to 2100 feet per minute as indicated on sheet 7 of MS35773. The heat rejected by the water and the heat gained by the air shall be separately calculated at each test condition. A performance tune shall be plotted as shown on MS35773, figure 1 or 2, as applicable. The heat rejection and core resistance values at 1500, 1800, and 2100 feet per minute air velocity shall be taken from the performance curve and recorded. The difference between the test data and standard core data shall be checked for conformance to 3.5.1.

4.5.2.1.6.2 Heat relection. alternate test method (b). When the same core section is used for more than one radiator type and size, water flow rates and air-velocity rates which overlap the entire range of required conditions shall be chosen for test points on the sample core, All the air velocities shall be tested at each chosen water flow rate. These results shall be plotted as shown on MS35773, figure 1 or 2, as applicable. The varying water flow rates in gallons per minute can be plotted as a parameter on these curves. From these tunes, a cross plot can be made with heat rejection in Btu/minute as the ordinate, water flow rates as the abscissa, and air velocity lines of 1500, 1800, and 2100 feet per minutes as the parameter. From this cross plot, the required values can be obtained and recorded. The data thus obtained shall be compared to the standard core values of MS35773, figures 1 and 2, as applicable. The difference between the test data and standard core data shall be checked for conformance to 3.5.1.

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4.5.2.1.7 Airflow. Airflow through the test core section shall be from either side.

4.5.2.1.8 Air-pressure-drop-corrections The air-pressure-drop measurements shall be corrected to standard conditions by use of the following formula:

$$\Delta P_C = \frac{(\Delta P)\rho}{\rho_0}$$

ΔP_C - Corrected drop, in H_2O.

ΔP - Measured drop, in H_2O.

ρ - Inlet air density, Lbm/cu. ft.

ρ_0 - Standard air density 0.075 Lbm/cu. ft.

4.5.2.1.9 Vivration. The radiator shall be filled with tapwater for the test specified in 4.5.2.4. The radiator shall be supported as specified in 4.5.2.1.1 and securely fastened to a rigid mounting bracket which shall be bolted to the vibration table to insure that the motion of the radiator shall be essentially the same as the motion of the platform. Means shall be provided for controlling the direction of vibration of the test machine and for adjusting and measuring frequencies and amplitudes of vibration to keep them within prescribed limits.

4.5.2.2 Heat rei ection and core resistance. The radiator or core, l s applicable, shall be tested as specified in 4.5.2.1.6 based on rated heat rejection as defined in 6.4.12. These values may be determined on l square-foot basis . These values shall than-be corrected to the area in square feet. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.3 Pressure cycling. The complete radiator shall be tested with all outlets closed. Pressure, variable from atmospheric to 150 percent of cap pressure not exceeding 18 psig, shall be applied at the inlet using air, steam or glycol and shall be maintained at a temperature of at least 212 'F during the test. When steam is used, means shall be provided to prevent the accumulation of water. The pressure cycling shall take place in 3 to 4 seconds and at a rate of 6 pressure cycles plus or minus 1 cycle per minute. The radiator shall be cycled to a minimum of 50,000 pressure cycles. The radiator shall be examined periodically for evidence of leakage or distortion. Only tube leaks and tube-to-header leaks, not to exceed three, shall be repaired or plugged before continuing the test. Nonconformance to 3.5.3 shall constitute failure of this test.

4.5.2.4 Vibration.

4.5.2.4.1 Resonance survey. The radiator which has been tested in accordance with 4.5.2.3 may be rebuilt or another radiator may be used for this test. The radiator shall be tested for leakage prior to start of this test. The radiator shall be prepared in accordance with 4.5.2.1.9 and vibrated at frequencies from 10 cycles per second(cps) to 33 cps at the table displacements specified in table I. The change in frequency shall be made at internals of 1 cps and maintained at each frequency for 1 minute. If resonance occurs at any point in the specified

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the specified range, the frequency of vibration at that point shall become the test condition for 4.5.2.4.2.

TABLE I. Table displacements.

Frequency range Cp s	Total table displacement inches
10 to 15	0.060 ± 0.006
16 to 25	0.050 ± 0.005
26 to 33	0.040 ± 0.005

4.5.2.4.2 Procedure. This test shall be run after completion of 4.5.2.4.1 if no leaks are observed. The radiator shall be vibrated for not less than 24 hours at the most critical resonant frequency. If no resonance was observed, this test shall be performed at 33 cps at the displacement referenced in table 1. The radiator shall be periodically examined during the test for evidence of seepage and leakage. Nonconformance to 3.5.4 shall constitute failure of this test.

4.5.2.5 Washing op ration. The radiator or core shall be filled with distilled water or tapwater of known pH index (see 6.4.13), at a temperature of 80 to 100 °F. The radiator shall then be shaken to assure thorough mixing and then allowed to stand for 1 hour. At the end of the l-hour period, 1/2 gallon of the contents shall be drained into a clean container and tested for pH change and chloride content. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Exanination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. One or more defects shall be cause for rejection.

111. Openings into radiators or replacement core assemblies not sealed as specified for level A or C (see S.1.1).
112. Gaskets not preserved as specified for level A or C (see 5.1.1 and 5.1.2).
113. Contents not immobilized within box as specified for level A or C (see 5.1.1 and 5.1.2).
114. Box closure and sealing not as specified for level A or C (see 5.1.1 and 5.1.2).

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115. Radiators or replacement core-assemblies of unlike description packed together for level A or B (see 5.2.1, 5.2.2, and 5.2.3).
116. Quantities packed together exceed the weight limitation of the box for level A or B (see 5.2.1 and 5.2.2).
117. Preservation or packing not in accordance with the referenced document as specified for level C (see 5.2.3),
118. Marking missing, illegible, incorrect, or incomplete for level A, B or C (see 5.3).

5. PACKAGING.

5.1 Presentation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), radiators or replacement core-assemblies shall not require application of a contact preservative. When specified (see 6.2), slush or flush interior of radiators replacement core-assemblies with VV-L-800 to ensure complete coverage and thoroughly drain excess. Openings into radiators or replacement core-assemblies shall be sealed with tape conforming to MIL-T-22085, type II or with caps or plugs of the appropriate size conforming to MIL-C-5501. The filler neck cap shall be secured in place to prevent loss. Gaskets for each replaceable core assembly shall be protected with fiberboard stiffeners and shall be preverved in accordance with 141L-P-116, method XC-1 or XC-3. Each radiator or each replaceable core-assembly with gaskets, shall then be placed in a close-fitting box conforming to PPP-B-636, class weather resistant, grade as applicable, style optiomal. The contents shall be blocked, braced or cushioned as applicable within the box to prevent movement or damage . Box closure and sealing shall be as specified for method V In the appendix to the box specification.

5.1.2 Level C. Each radiator or replacement core-assembly with gaskets, shall be preserved as specified for level A except the boxes shall be domestic class.

5.2 Packing. Packing shall be level A, level B or C as specified (sea 6.2).

5.2.1 Level A. Radiators of like description or replacement core-assemiblies of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601 overseas type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Radiators or replacement core-assemblies with gaskets shall be packed as specified in 5.2.1 for level A except that boxes shall be domestic type and strapping may be finish A.

5.2.3 Level C. Radiators or replacement core-assemblies with gaskets, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

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5.3 Marking. In addition to any special marking specified in the contract or purchase description (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The radiators covered by this specification are intended for use with liquid-cooled internal combustion engines to dissipate the heat from the coolant.

6.2 Accuisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of radiator or core, when applicable, required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is not required (see 3.2, 4.3 and 6.3).
e. Level of prservation and packing required (see 5.1 and 5.2).
f. When interior of radiator or replacement core-assemblies are to be preserved (see 5.1.1).
g. Any special marking (see 5.3).

6.3 First article. inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions apply to the text of this specification.

6.4.1 Rated internal pressure. The rated internal pressure is 7 psig for radiators conforming to MS35773-1 through -11.

6.4.2 standard air Standard air is air at a temperature of 70 “F, a barometric pressure of 29.92 inches of mercury, and a density of 0.075 pound per cubic foot.

6.4.3 Rated air velocity. The rated air velocities are those listed on sheet 7 of MS35773.

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6.4.4 Rated coolant flow. The rated coolant flow is the flow listed on sheet 7 of MS35773, and is based on approximately a 10 “F drop in coolant temperature in the radiator.

6.4.5 Heat transfer - heat energy absorbed by air flow. The heat energy absorbed by air flow is the product of the air flow, specific heat of air, and air temperature rise.

Heat energy absorbed by air flow - $wc_p \Delta T$

w - air flow rate, Lbm/Min.

c_p - 0.24 Btu/Lbm/°F

ΔT - air temp rise, “F

6.4.6 Heat transfer - the heat rejection from water flow. The heat rejection from water flow is the product of the coolant flow, specific heat of coolant and coolant temperature drop.

Heat transfer (heat rejection from water flow) - wc_p A_T

w - water flow rate, Lbm/Min.

c_p - 1 Btu/Lbm/°F

A_T - water temp drop, ‘F

6.4.7 Specific heat at constant pressure of water. The specific heat at constant pressure of water is based on the average water temperature (see 6.4.9) and have the value of 1 Btu/lb./°F for water temperatures up to 215 ‘F.

6.4.8 Specific heat at constant pressure of air. The specific heat at constant pressure of air shall have the value of 0.24 Btu/lb./°F for air temperatures up to 215 “F and pressure around 1 (atm.).

6.4.9 Average water temperature. The average water temperature of the radiator is the sum of the water inlet and outlet temperatures divided by 2.

6.4.10 Rated potential. Rated potential is the average water temperature minus the entering air temperature and have a value of 80 oF.

6.4.11 Observed potential. Observed potential is the observed average water temperature minus the observed entering air temperature during heat-rejection and core-resistance test.

6.4.12 Rated heat rejection. The rated heat rejection is the total observed heat rejection in Btu’s per minute multiplied by the rated potential and divided by the observed potential.

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6.4.13 Index of acid intensity. The pH index is the negative logarithm of the hydrogen ion concentration per liter. A pH index of 7 is a neutral solution. A pH index of less than 7 is an acidic solution. A pH index of greater than 7 is a basic solution.

6.4.14 Resonance. Resonance is a condition of maximum magnification of an applied vibration. It is usually manifested by visibly increased vibration of the radiator under test.

6.5 Subject term(key word) listing.

Bolted-tank radiator
Coolant
Engine, internal combustion
Liquid-cooled
Radiator core
Solder-tank radiator

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ME
Air Force - 99

Review activities:

Army - AT
Air Force - 82
DLA - CS

Preparing activity:

Army-Me

Project 2930-0005

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-453060	2. DOCUMENT DATE (YYMMDD) 911015

3. DOCUMENT TITLE
Radiators, Engine Cooling, Industrial

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1806	(2) AUTOVON 354-1806
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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